

P3 — Frontend + Product + Geospatial UI

SIH 26143 · Detailed 6-Phase Work Plan · 6 owned work packages

ROLE OBJECTIVE

Own the investigator-facing geospatial product and make every important evidence layer understandable on one screen.

HOW TO USE THIS PLAN

Each package has four parts: what to build/do, the concrete deliverable, dependencies, and an acceptance condition. The owner is responsible for making the output usable by the next phase—not merely completing a task locally.

PHASE 1 — DATA & PREPROCESSING

Work package 1: Data-to-UI specification

Map every backend field to a visible dashboard element: scene metadata, spill mask, confidence, source corridor, origin window, drift particles, vessel track, evidence family and uncertainty; define empty/error states.

Deliverable: UI data contract + screen/state specification.

Dependencies / acceptance: Depends on P2/P1/P4/P5 schemas. Acceptance: no UI field is invented or disconnected from a backend source.

PHASE 2 — SPILL DETECTION

Work package 2: Detection visualisation

Build SAR scene view with spill-mask overlay, legend, confidence display and incident metadata; add controls for original imagery versus model output; make clear which layers are observed and which are model-generated.

Deliverable: functional detection map component.

Dependencies / acceptance: Depends on P1 outputs and MapLibre layer setup. Acceptance: user can distinguish SAR observation from predicted mask.

PHASE 3 — SOURCE RECONSTRUCTION

Work package 3: Source/drift visualisation

Display detected slick, backward particles, source corridor, origin-time window and forward drift corridor; add uncertainty styling and explanatory labels; avoid presenting the reconstructed source as an exact point.

Deliverable: reconstruction map component.

Dependencies / acceptance: Depends on P5 output contract. Acceptance: uncertainty and time-window information remain visible at all relevant zoom levels.

PHASE 4 — AIS CORRELATION

Work package 4: Vessel-track UI

Render historical AIS tracks and positions around the reconstructed source region; show vessel metadata, time filtering, AIS-gap indicators and track continuity; support toggling evidence layers.

Deliverable: AIS trajectory map layer + controls.

Dependencies / acceptance: Depends on P4 trajectory output and P2 query API. Acceptance: candidate tracks are synchronized to the incident time window.

PHASE 5 — EVIDENCE FUSION + UI

Work package 5: Investigator dashboard

Assemble the full investigation view: SAR, spill polygon, source corridor, drift, AIS tracks, candidate list, evidence cards and timeline; label synthetic/demo content explicitly when applicable.

Deliverable: integrated investigator dashboard.

Dependencies / acceptance: Depends on all core APIs. Acceptance: one incident can be followed from detection to candidate evidence without switching tools.

PHASE 6 — VALIDATION & DEPLOYMENT

Work package 6: End-to-end UI integration

Replace static placeholders where real outputs exist; add loading, error, insufficient-evidence and no-data states; verify every number shown on the screen is sourced from the backend or clearly labelled illustrative.

Deliverable: presentation-ready working UI connected to the actual pipeline.

Dependencies / acceptance: Depends on integrated backend. Acceptance: demo scenario runs without hidden/manual data edits.

CROSS-TEAM HANDOFF RULE

No component is considered complete if its output cannot be consumed by the next owner. Preserve IDs, timestamps, coordinate systems, provenance, uncertainty and failure states across every handoff.

Core project rule: AIS gaps, uncertain metocean forcing, ambiguous SAR signatures or weak evidence must increase uncertainty or trigger abstention—not be converted into fabricated confidence.